```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Analog Clock</title>
    <style>
        /* ============================== */
        /* CSS STYLING (UPDATED) */
        /* ============================== */

        body {
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
            margin: 0;
            background-color: #f4f4f9;
            font-family: Arial, sans-serif;
        }

        .clock {
            width: 350px;
            height: 350px;
            border: 10px solid #333;
            border-radius: 50%;
            position: relative; /* Crucial for positioning the hands */
            background: #fff;
        }

        .hand {
            /* CRITICAL FIXES START HERE */
            position: absolute;
            /* Setting top/left to 50% and transforming back by half its own size
            centers it perfectly. */
            top: 50%;
            left: 50%;
            transform-origin: center center; /* Set rotation origin to the element's
            defined center */
            margin-top: -1px; /* Adjust for border effects */

            /* This combination ensures the rotational point is exactly in the middle
            of .clock */
        }


        /* The Hour Hand (Short and thick) */
        #hour-hand {
            width: 8%; /* Slightly wider to look balanced */
            height: 70px; /* Length - shortened slightly for visual balance */
            background-color: #333;
            transform: translate(-50%, 0); /* Adjusting offset after positioning
            logic is complex, simple translation of origin helps */
            /* We rely on the general hand styling + absolute positioning to handle
            centering. */
        }

        /* The Minute Hand (Medium length) */
        #minute-hand {
            width: 4%;
            height: 130px; /* Length */
            background-color: #333;
             transform: translate(-50%, 0);
        }

        /* The Second Hand (Thin and distinct) */
        #second-hand {
            width: 2px;
            height: 160px; /* Length */
            background-color: red;
            transform: translate(-50%, 0);
        }
```

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68
69
70          /* Center Pivot remains the same - it's correct for a visual overlay */
71          .clock::before {
72              content: '';
73              position: absolute;
74              top: 50%;
75              left: 50%;
76              transform: translate(-50%, -50%);
77              width: 12px;
78              height: 12px;
79              background-color: #333;
80              border-radius: 50%;
81              z-index: 40;
82          }
83
84      </style>
85  </head>
86  <body >
87
88      <div class="clock">
89          <!-- Note on the Hand Styling Trick: -->
90          <!-- Because we are using `position: absolute` and want rotation to happen at
            50%,
91               we define the hand's dimensions (height/width) and then use
                 'translate(-50%, 0)'
92               in CSS for a visual offset, but JS must still rotate relative to the
                 central pivot. -->
93          <div id="hour-hand" class="hand"></div>
94          <div id="minute-hand" class="hand"></div>
95          <div id="second-hand" class="hand"></div>
96      </div >
97
98      <script>
99          /* ============================== */
100         /* JAVASCRIPT LOGIC (NO CHANGES NEEDED) */
101         /* ============================== */
102
103         const hourHand = document.getElementById('hour-hand');
104         const minuteHand = document.getElementById('minute-hand');
105         const secondHand = document.getElementById('second-hand');
106
107         function updateClock() {
108             // Get the current local time
109             const now = new Date();
110             let hours = now.getHours();
111             let minutes = now.getMinutes();
112             let seconds = now.getSeconds();
113
114             // Normalize to 12-hour format
115             hours %= 12;
116             minutes %= 60;
117             seconds %= 60;
118
119             /* Calculate Degrees */
120
121             // Second Hand: 360 degrees in 60 seconds (6 deg/sec)
122             let secondDegrees = (seconds * 6);
123
124             // Minute Hand: 360 degrees in 60 minutes. Added bonus from seconds.
125             let minuteDegrees = ((minutes * 6) + (seconds * 0.1));
126
127             // Hour Hand: 360 degrees in 12 hours. Bonus from minutes and seconds.
128             let hourDegrees = ((hours * 30) + (minutes * 0.5) + (seconds * 0.0083));
129
130
131             /* Apply Rotation */
132
133             secondHand.style.transform = `rotate(${Math.round(secondDegrees)}deg)`;
134             minuteHand.style.transform = `rotate(${Math.round(minuteDegrees)}deg)`;
135             hourHand.style.transform = `rotate(${Math.round(hourDegrees)}deg)`;
136         }
```

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137
138             // Initial call and continuous update (50ms for smoothness)
139             updateClock();
140             setInterval(updateClock, 50);
141         </script>
142     </body>
143     </html>
```